

# Creating a Dockerfile

## Title

## Objective

The objective of this task was to learn how to create a Dockerfile and use it to build a Docker image for containerizing an application.

## Introduction

A Dockerfile is a text file that contains a set of instructions used to build a Docker image automatically. It defines the application's environment, dependencies, source code, and startup command. Dockerfiles help developers create consistent and portable environments, making applications easier to deploy across different systems.

## Requirements

- Docker installed on the system
- A sample application (Java, Python, Node.js, etc.)
- Command-line terminal
- Text editor (Visual Studio Code or any editor)

## Sample Dockerfile

```
FROM openjdk:17-jdk-slim

WORKDIR /app

COPY . .

RUN javac Hello.java

CMD ["java", "Hello"]
```

## Explanation of Dockerfile Instructions

- **FROM** – Specifies the base image used to build the application.
- **WORKDIR** – Sets the working directory inside the container.
- **COPY** – Copies the application files from the local system into the container.
- **RUN** – Executes commands while building the image.
- **CMD** – Specifies the default command to run when the container starts.

## Steps Performed

1. Created a project directory.
2. Created a Dockerfile in the project folder.
3. Added the required Docker instructions.
4. Saved the Dockerfile.

5. Built the Docker image using:

```
docker build -t my-java-app .
```

6. Verified that the image was created:

```
docker images
```

7. Ran the Docker container:

```
docker run my-java-app
```

8. Verified that the application executed successfully inside the container.

## Observation

- The Docker image was created successfully.
- The container started without errors.
- The application executed correctly in an isolated environment.

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Jul 20 06:34:32 2026 from 18.206.107.28
ubuntu@ip-172-31-65-154:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-65-154:~$ cd /
ubuntu@ip-172-31-65-154:/$ ls
bin  boot  dev  etc  home  lib  lib64  lost+found  media  mnt  opt  proc  root  run  sbin  snap  sr
ubuntu@ip-172-31-65-154:/$ cd opt
ubuntu@ip-172-31-65-154:/opt$ pwd
/opt
ubuntu@ip-172-31-65-154:/opt$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
70 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-65-154:/opt$ sudo apt install apache2 -y
apache2 is already the newest version (2.4.66-2ubuntu2.4).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 70
ubuntu@ip-172-31-65-154:/opt$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
70 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

**i-071479beb0dcc12a3 (docker-server)**  
PublicIPs: 32.197.13.173 PrivateIPs: 172.31.65.154

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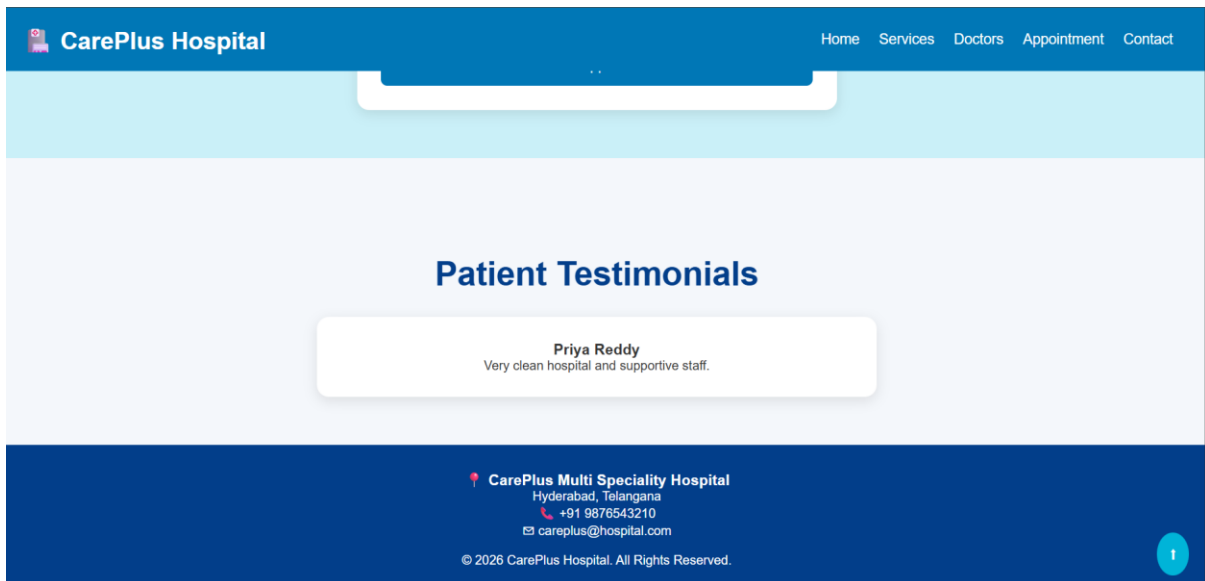
Phone Number

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## Advantages of Dockerfile

- Automates application deployment.
- Ensures consistent environments across systems.
- Reduces configuration errors.
- Simplifies application sharing and deployment.
- Supports version control for infrastructure.

## Learning Outcome

- Understood the purpose of a Dockerfile.
- Learned the basic Dockerfile instructions.
- Gained experience in building Docker images.
- Learned how to run applications inside Docker containers.
- Improved understanding of containerization concepts.

## Conclusion

The Dockerfile was created successfully, and the Docker image was built and executed without issues. This activity provided practical knowledge of containerization and demonstrated how Docker simplifies application deployment by packaging code, dependencies, and runtime into a single portable image.